Suviksha Viswanath Biranthabail

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Summary

Computer Science student specializing in cybersecurity, with experience building and analyzing systems across web, data, and security domains. Skilled in identifying vulnerabilities using OWASP practices, developing lightweight security tools, and working with real-world datasets. Strong foundation in software engineering, system design, and problem-solving, with proficiency in Python and Java.

Education

RV University 2023 – 2027
B.Tech (Hons) Computer Science – Cybersecurity Major, FinTech Minor *Bangalore, India*

- **Relevant Coursework:** Data Structures, Algorithms, Computer Networks, Database Systems, Machine Learning, Cybersecurity, Web Technologies
- **Network Security Lab (Hardware):** Hands-on lab using Ruckus ICX7150 switches and routers covering NAT, TCP socket programming (plaintext and OpenSSL-encrypted), Telnet-based device management, VLAN construction and inter-VLAN routing, Access Control Lists, Router-on-a-Stick, OSPF, DHCP snooping, and port security.
- **Embedded Systems and Edge AI Lab:** Worked with Raspberry Pi Pico RP2040 and ESP32 microcontrollers to implement binary LED displays, assembly-to-C integration, dual-core task execution, IPC, and FreeRTOS message queues.
- **TinyML and Edge Inference:** Built TensorFlow models for sine wave generation and MNIST digit recognition, converted models to TensorFlow Lite, and deployed inference on ESP32 devices.
- **IoT and Systems Programming:** Developed MQTT-based ESP32 communication for remote LED control and explored low-level C concepts including volatile variables, structures, memory padding, and pointer operations.

Experience

Grant Thornton Indus – US Audit Support Sep 2026 – Dec 2026
SASS Intern, Audit & Security Assurance *Bengaluru, India*

- Support SOC 1/SOC 2 engagements by testing security controls such as logical access, change management, and incident response against client environments
- Review and validate client-submitted evidence (access reviews, configurations, system logs) to assess control design and operating effectiveness, and document exceptions
- Contribute to SOC report preparation by mapping evidence to Trust Services Criteria and drafting test results for review
- Collaborate daily with the US engagement team on deliverables, client queries, and timelines

Amar Seva Sangam (USA Chapter) Dec 2025 – January 2026
Web Developer (Freelance) *Remote*

- Developed and deployed a production-ready single-page website hosted on Vercel
- Integrated Zeffy payment workflows enabling seamless online fundraising
- Built dynamic contact systems with automated email routing
- Continuously improved UI/UX and site responsiveness through iterative updates

RV University May 2024 – Dec 2024
Cybersecurity Intern *Bangalore, India*

- Performed vulnerability assessments using OWASP methodologies across web and mobile systems
- Developed UI components for internal security dashboards and analysis tools
- Built a mobile APK penetration testing tool for automated security reporting
- Collaborated in agile workflows for risk identification and mitigation

RV University June 2024 – August 2024
AI for Cybersecurity – Student Research *Bangalore, India*
- Analyzed large-scale security datasets to identify anomalies and suspicious patterns
- Performed exploratory data analysis on security logs and network data
- Supported ML-based security insights through data preprocessing and feature analysis

Fillum – Short Film Platform May 2025 – July 2025
Frontend Developer Intern (On Hold) *Remote*
- Developed responsive UI components using React, HTML, CSS, and JavaScript
- Designed user-facing interfaces with focus on usability and clean design
- Improved frontend modularity and component consistency

Arc22 Ventures July 2025 – September 2025
Backend & Web Operations Intern (On Hold) *Remote*
- Engineered backend workflows using Wix Studio with dynamic datasets
- Structured event registration systems and tournament listing logic
- Supported dashboard development and data pipelines

Projects

Vaani Rekha – Multilingual AI Scam Call Detection System
- Built a near real-time system to detect AI-generated and human scam voice calls across multiple Indian languages.
- Implemented wav2vec2 XLS-R, a cross-lingual speech model, to classify call audio as AI-generated or human-spoken.
- Integrated multilingual speech-to-text pipelines to transcribe call audio for downstream fraud-pattern analysis.
- Designed detection logic to flag high-risk scam patterns such as OTP requests and fake banking alerts.
- **Top 2% of 40,000+ participants – India AI Buildathon 2026**

IoT Network Security Dashboard
- Built a three-tier network monitoring system with a Python scanner, a Node.js/Express + MongoDB backend, and a React dashboard.
- Implemented automatic device discovery via ARP scanning to detect new, unknown, and rogue devices on the network.
- Developed rogue device quarantine workflows that enforce access blocks using Windows Firewall rules from the scanner.
- Built a live dashboard surfacing device counts, alerts, and quarantine status through REST endpoints across the stack.

MobiPent – Mobile APK Security Analyzer
- Built a full-stack mobile security platform with a FastAPI backend and Expo React Native frontend for OWASP MASVS-aligned APK analysis.
- Implemented JWT-based authentication with token-protected endpoints and SQLite-backed user management.
- Developed APK analysis modules using Androguard for permission analysis, secret detection, and cryptography review.
- Built individual tool scans and a full OWASP MASVS compliance scan returning structured, actionable security reports.

Darknet Dataset Anomaly Detection
- Processed large-scale darknet traffic datasets to identify anomalies, suspicious behavioral patterns, and potential threat indicators
- Performed exploratory data analysis (EDA) and feature engineering to extract meaningful signals for ML-based anomaly detection models
- Applied preprocessing techniques including normalization, class imbalance handling, and dimensionality reduction to improve model accuracy
- Evaluated detection performance using precision, recall, and F1 metrics to provide interpretable insights into darknet traffic behavior

Handwritten Equation Solver (CNN)

- Built a Convolutional Neural Network (CNN) model to recognize and classify handwritten mathematical symbols from scanned inputs
- Converted recognized symbols into structured mathematical expressions and computed results using an expression parser
- Trained and validated the model on a labeled dataset of handwritten symbols, applying data augmentation to improve generalization
- Designed a preprocessing pipeline for image normalization and noise reduction to enhance symbol detection accuracy

ApexZone – Fitness Web Application

- Developed a responsive fitness web application using React, HTML, CSS, and JavaScript with a focus on clean UI and smooth user experience
- Built interactive dashboards with personalized workout modules, exercise tracking, and dynamic content rendering
- Implemented modular React component architecture for maintainability and reusability across workout categories and user views
- Ensured cross-device responsiveness and accessibility through adaptive layouts and performance-optimized rendering

VLAN-Based Secure Exam Network

- Designed and deployed a secure dual-VLAN wireless network using Ruckus Networks hardware, including a Layer 3 switch and enterprise access points, to isolate exam traffic from regular campus traffic
- Configured VLAN segmentation, inter-VLAN routing, DHCP pools, and Access Control Lists (ACLs) on a Ruckus ICX 7150-C12P switch to enforce secure communication policies
- Implemented separate SSIDs for normal users and exam participants, ensuring exam clients could only access the dedicated exam server while blocking unauthorized internet and cross-network access
- Simulated a real-world university secure examination environment with automated IP allocation, Layer 2/Layer 3 isolation, and network access control mechanisms

Brick Breaker Game (Java OOP Mini Project)

- Developed a multi-level 2D Brick Breaker game in Java applying OOP principles – abstraction, inheritance, polymorphism, and encapsulation – for clean and maintainable architecture
- Implemented interactive gameplay mechanics including paddle control, ball physics, collision detection, a scoring system, and progressively increasing difficulty levels
- Designed dynamic brick layouts with moving bricks in advanced levels using a dedicated `BrickMap` class for modular and scalable game management
- Added sound effects, win/lose game states, and real-time rendering using Java Swing/AWT; structured the codebase with reusable classes and interfaces for extensibility

Technical Skills

Languages: Python, Java, C, JavaScript
Frameworks: React.js, TensorFlow, React Native
Technologies: Machine Learning, Data Analysis, Web Development
Security: OWASP, Network Security, Vulnerability Assessment, APK Analysis
Tools: Wireshark, Nmap, Metasploit, Nessus, Kali Linux
Concepts: Data Structures, System Design, Networking

Certifications

Affective Computing (88/100) — NPTEL (SWAYAM)
Emotion recognition, HCI, affect-aware intelligent systems.

Design & Implementation of Human-Computer Interfaces (64/100) — NPTEL (SWAYAM)
User-centered design, usability engineering, interface evaluation.

Programming in Modern C++ (55/100) — NPTEL (SWAYAM)
Modern C++, STL, templates, OOP, memory management.

Getting Started with Cisco Packet Tracer — Cisco Networking Academy
Network simulation, routing, switching, topology configuration, troubleshooting.

Network Security — IBM SkillsBuild
Cybersecurity fundamentals, threat identification, access control, security policies, risk mitigation.

Big Data 101 / Hadoop 101 / Spark Fundamentals I — IBM Skills Network
HDFS, MapReduce, Apache Spark, distributed data processing.

Computer Networking Basics: How Computers Communicate — IBM SkillsBuild
TCP/IP, OSI model, communication protocols, data transmission.

ISC2 Certified in Cybersecurity (CC) Exam Passed — ISC2
Security principles, network security, access control, risk management, incident response.